Installing OpenCV for Python

To install the OpenCV library, simply open your command prompt or terminal window and run the following command:

```
pip install opencv-python
```

Step 1: Import the OpenCV Package

```
import cv2

imagePath = 'input_image.jpg'
```

Step 2: Read the Image

```
img = cv2.imread(imagePath)
```

Step 3: Convert the Image to Grayscale

```
gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

Step 4: Load the Classifier

```
face_classifier = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)
```

Step 5: Perform the Face Detection

```
face = face_classifier.detectMultiScale(
    gray_image, scaleFactor=1.1, minNeighbors=5, minSize=(40, 40)
)
```

Step 6: Drawing a Bounding Box

```
for (x, y, w, h) in face:
    cv2.rectangle(img, (x, y), (x + w, y + h), (0, 255, 0), 4)
```

Step 7: Displaying the Image

To display the image with the detected faces, we first need to convert the image from the BGR format to RGB:

```
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

Now, let's use the Matplotlib library to display the image:

```
import matplotlib.pyplot as plt
plt.figure(figsize=(20,10))
plt.imshow(img_rgb)
plt.axis('off')
```

Python code for face Dectection

```
pip install opencv-python
import cv2
import matplotlib.pyplot as plt

imagePath = 'input_image.jpg'
img = cv2.imread(imagePath)
gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
face_classifier = cv2.CascadeClassifier(
    cv2.data.haarcascades + "haarcascade_frontalface_default.xml"
)
face = face_classifier.detectMultiScale(
    gray_image, scaleFactor=1.1, minNeighbors=5, minSize=(40, 40)
)
for (x, y, w, h) in face:
    cv2.rectangle(img, (x, y), (x + w, y + h), (0, 255, 0), 4)
img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
```

plt.figure(figsize=(20,10))

plt.imshow(img_rgb)

plt.axis('off')

Prediction Model using XGBoost

```
from sklearn import datasets
from sklearn import metrics
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('ggplot')

import lightgbm as ltb
```

```
datapath ="/content/drive/MyDrive/Cardio/FinalFeature.csv"
Labelpath ="/content/drive/MyDrive/Cardio/XGLab.csv"


data = pd.read_csv(datapath)
#data = data1.transpose()
Label = pd.read_csv(Labelpath)
#label=label1.transpose()
print(data)

X_train,X_test , y_train, y_test =
train_test_split(data,Label,test_size=0.20)




!pip3 install xgboost
from xgboost import XGBClassifier
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score
from sklearn.datasets import load_wine
model = XGBClassifier(use_label_encoder=False, eval_metric='mlogloss')
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
accuracy
```